



Team Fih

September - 20- 2026

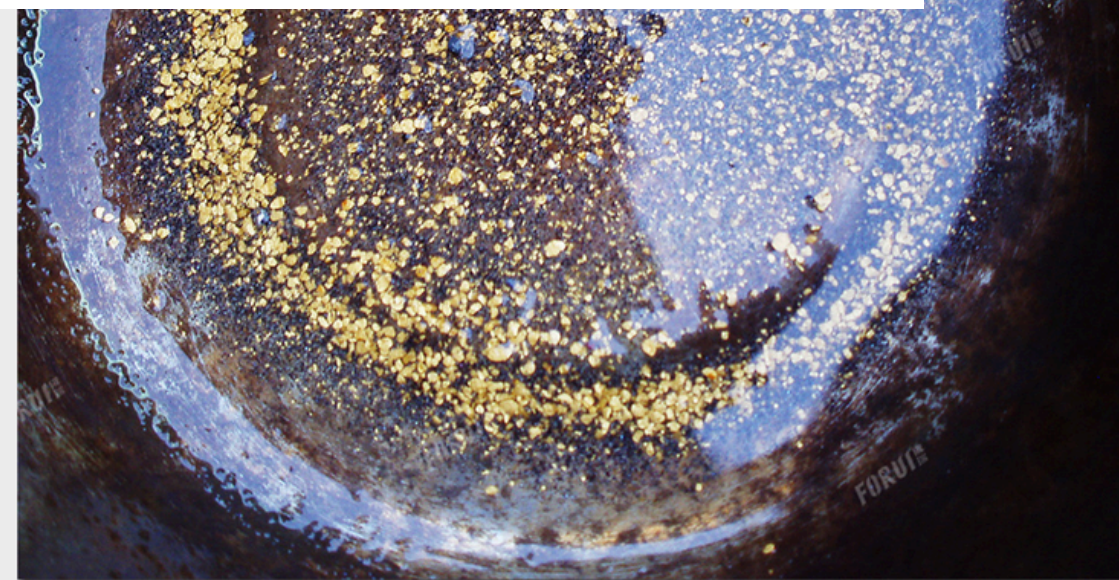
TRACEGUARD

Privacy is the foundation, not an obstacle



Presented By :
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DATA IS A GOLDMINE





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WHAT THE PII ?!

Personally Identifiable Information or Personal Data*, any information that can be used to identify a specific individual

Direct Identifier

Immediate identification
e.g. msisdn, imsi, imei**

Quasi Identifier

Alone harmless, but combining multiple can lead to identification.
e.g. time_start, enb, province, radio_access_type, imei**

Sensitive Attribute

Outliers can identify individuals
application_category, data_GB_sum, im_video_GB_sum, im_audio_GB_sum, tethering_data_GB_dl_sum

Behavioural

Can be used in large quantities and combined with external data to track behaviours.
tp_dl_avg, tp_ul_avg,
tp_dl_filtered_avg, cont_rtt *,
initial_rtt_radio_avg, tcp_retrans *,
http_response_time_avg, http_sr_avg

* GDPR (esp. Art 4); Recital 26, CJEU C-413/23 P EDPS v SRB (2025), WP29 Opinion 05/2014 on Anonymisation Techniques, e-Privacy Directive (Directive 2002/58/EC) Article 9(1), EDPB Guidelines 02/2026 on Anonymisation



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Guiding Star : Minimize data loss

WHO IS ANON?

Columnal :

- Drop direct IDS
- IMEI → TAC, device model is informative, not defining (QI)
- Time → hours, harder to track

Row-level:

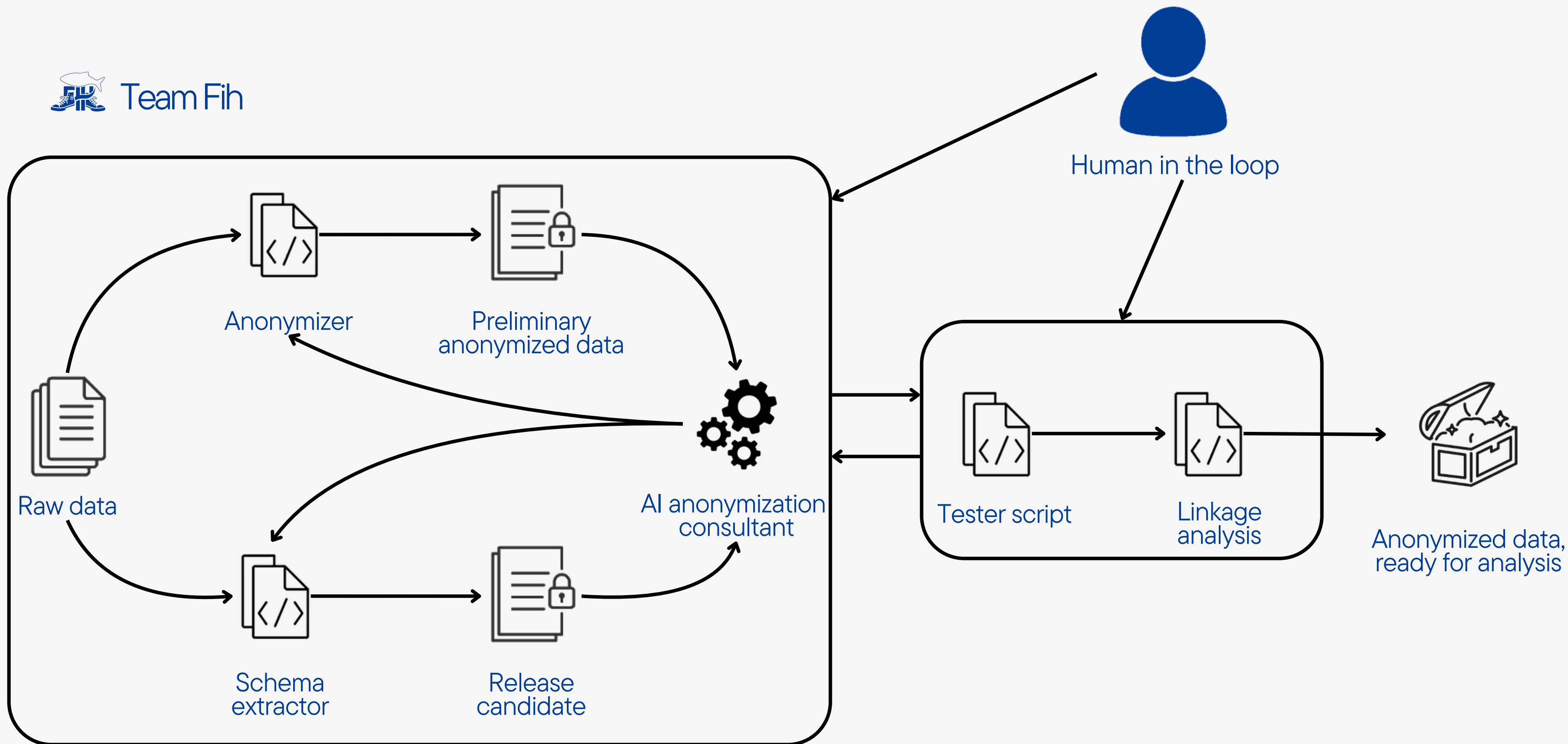
- QIs iterative anonymization (2 step, check against a k-anonymity test)
- max linkage is 4 (de Montjoye, Yves-Alexandre, et al. Unique in the crowd: The privacy bounds of human mobility. Scientific reports, 2013, 3:1:1376.)
- TAC randomization, same results



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RISKS & REWARDS

Risks	Reward	Level	Reason
8 combined data singeling	Higher data retention	Low	Requires external dataset or insight to identify a person
IMEI → TAC transformation	Gives partial, highly insightful information about devices	Medium	Can lead to fingerprinting
Telecom data	usage metrics, helps development of services in areas	Medium	Highly behavioural, behaviour fingerprints persist across mutiple days
AI	Reviews anonymization, gives second PoV	Low	AI act reporting



PIPELINE



METRICS

Data utility scorecard - anonymised vs original

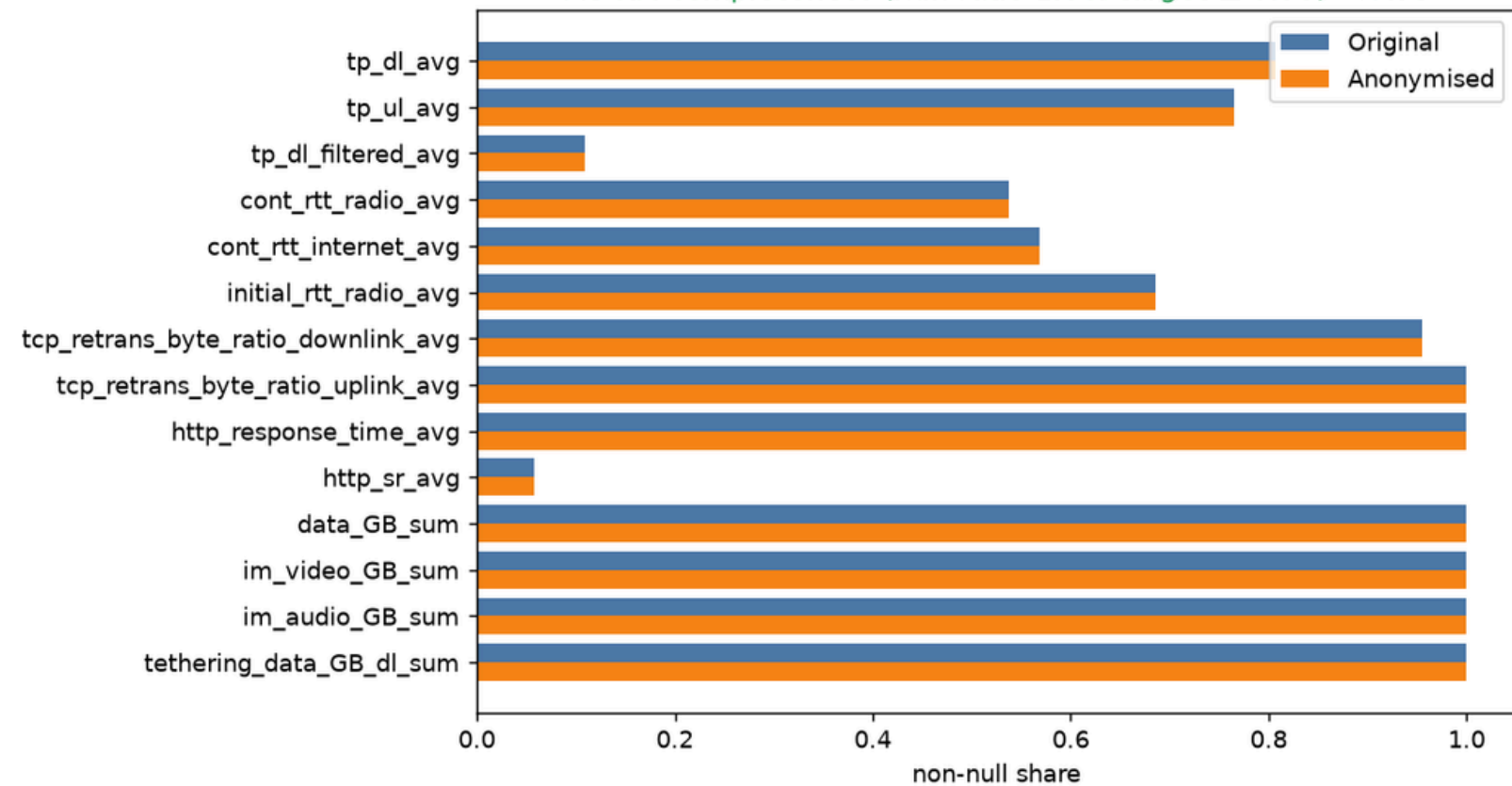
	Metric	Baseline	Target	Result	Status
M1	Row retention	100%	$\geq 95\%$	100.0%	PASS
M3	KPI completeness	raw non-null rate	ratio ≥ 0.95 per column	min ratio 1.00	PASS
M4	Distribution fidelity	KS = 0	KS ≤ 0.05 , med/P95 err $\leq 5\%$	max KS 0.002 (cont_rtt_internet_avg)	FAIL
M5	Categorical fidelity	JSD = 0	JSD ≤ 0.02	max JSD 0.0767	FAIL
M6	Correlation preservation	$\Delta\rho = 0$	max $ \Delta\rho \leq 0.05$	max $ \Delta\rho $ 0.024	PASS
M7	Known-finding reproduction	raw values	RAT gap within $\pm 5\%$, province $\rho \geq 0.95$	gap err 0.0%, ρ 1.00	PASS
M8	Downstream ML utility (TSTR)	orig-trained R^2 0.74, F1 0.78	$R^2 \geq 95\%$, F1 $\geq 90\%$ of baseline	R^2 101%, F1 100%	PASS
M9	Cell degradation detection	100% recall	recall $\geq 90\%$	recall 99% (min of 2 signals)	PASS



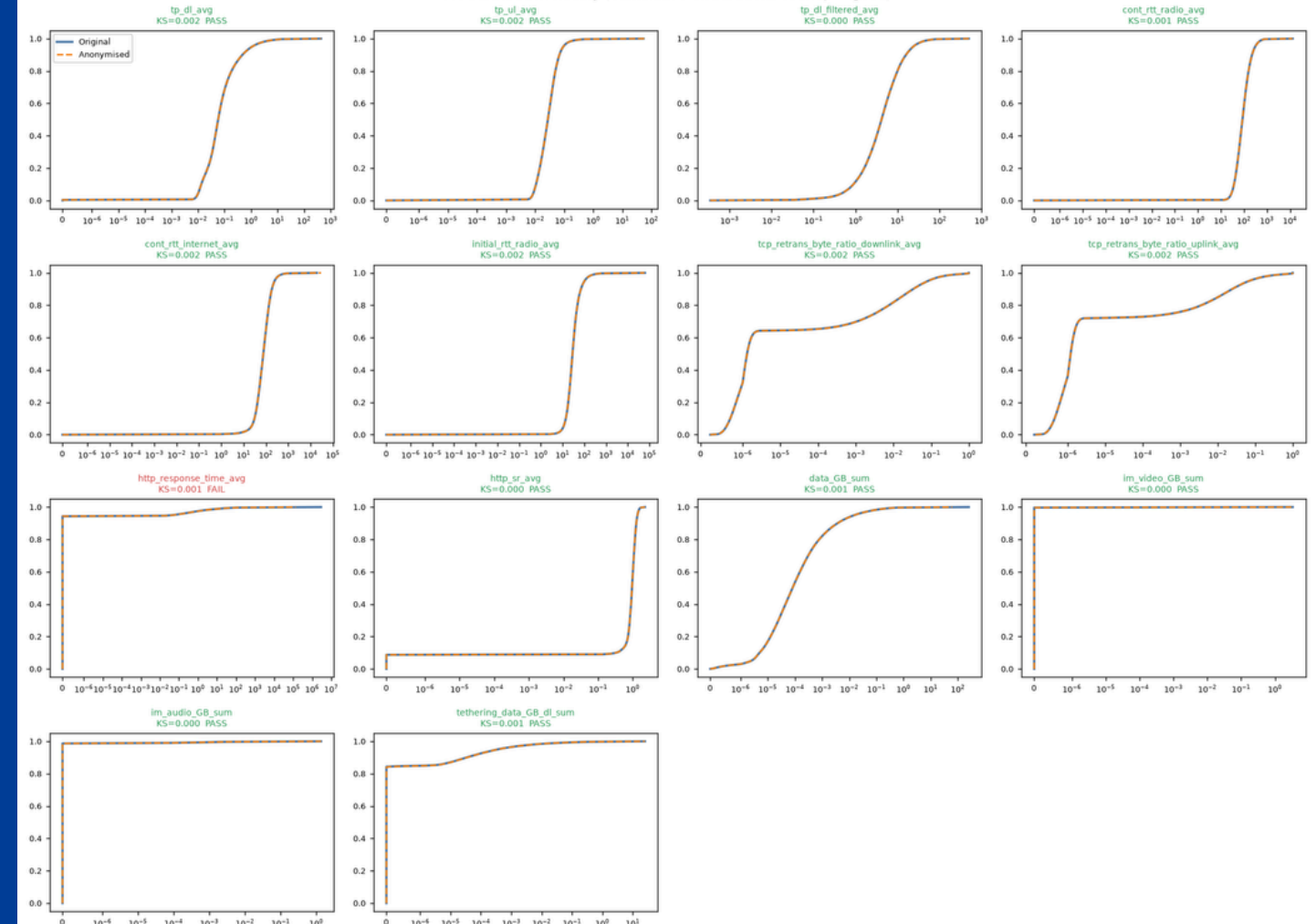
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ANALYTICAL POTENTIAL

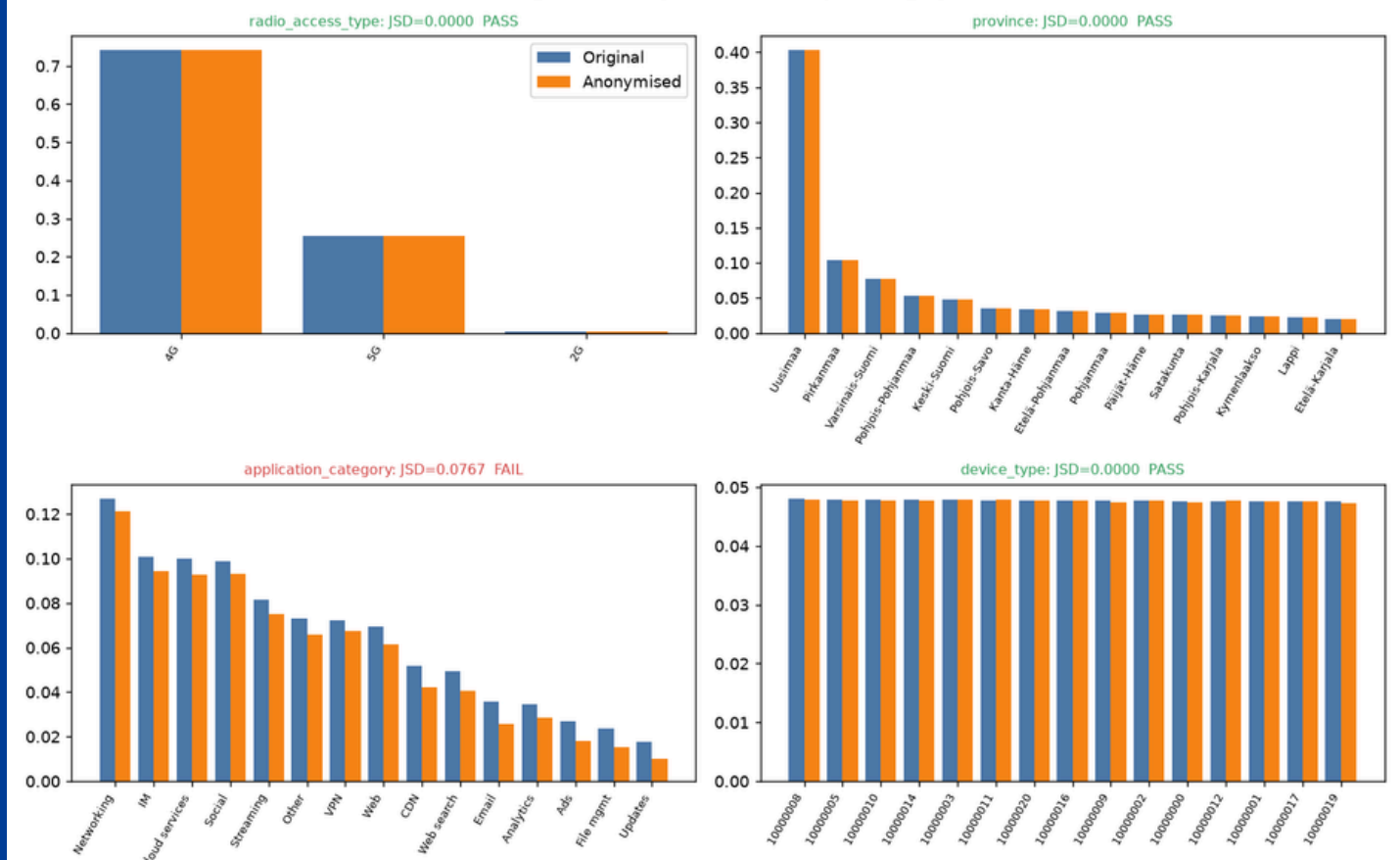
M3 KPI completeness (min ratio 1.00, target ≥ 0.95) - PASS



M4 Distribution fidelity (cumulative distributions overlap = good)



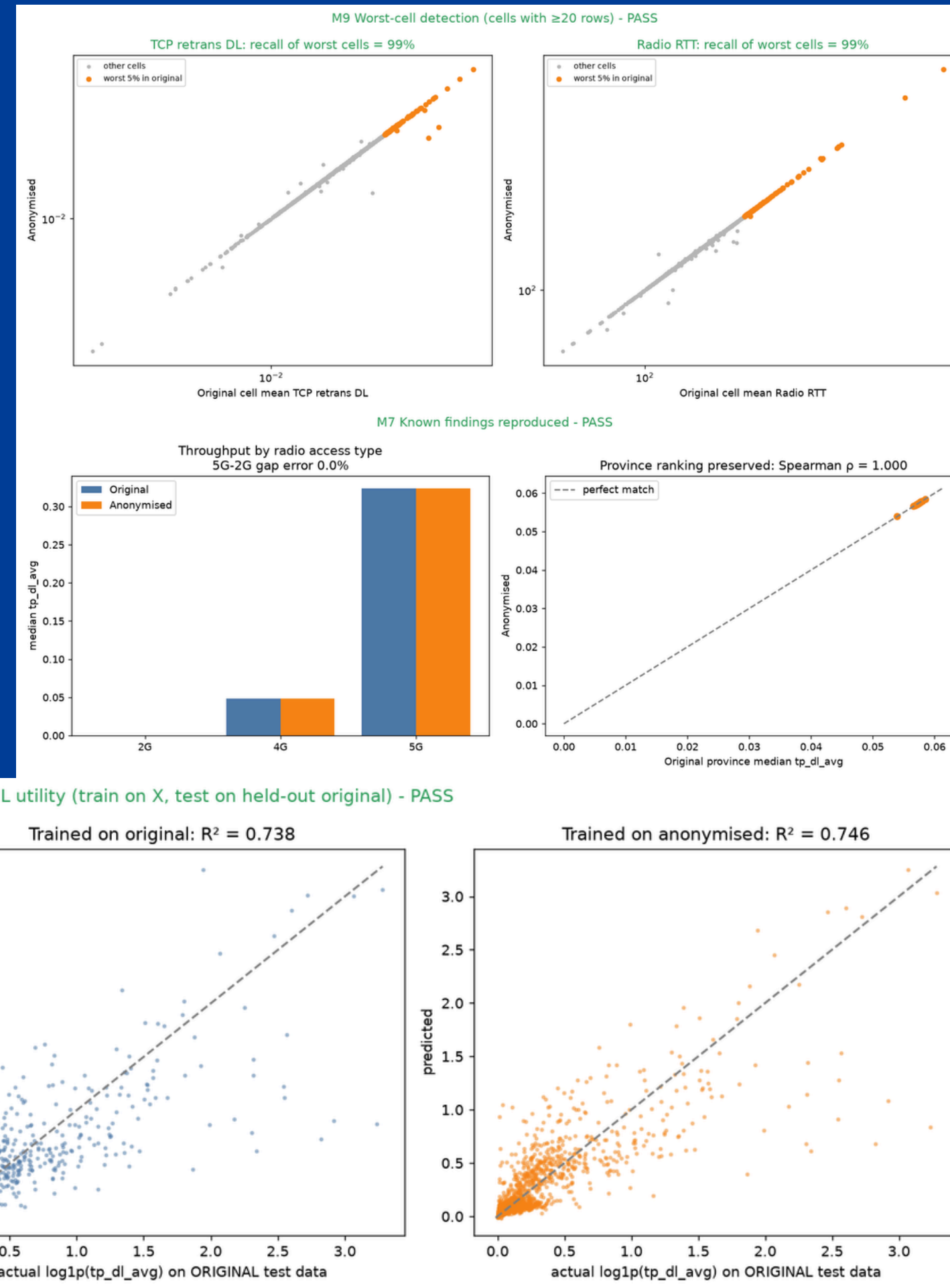
M5 Categorical fidelity (share of rows per category)





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ANALYTICAL CAPABILITY





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- 1.- GDPR
- 2.- Data Act
- 3.- ePrivacy Act
- 4.- AI Act
- 5.- WP29 Opinion 05/2014 (WP216)
- 6.- Electronic Communications Services Act (917/2014)
- 7.- Strong electronic identification act (617/2009)
- 8.- Cybersecurity Act (124/2025)
- 9.- EDPB Guidelines 02/2026 on Anonymisation

COMPLIANCE



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THANK YOU

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